

# Maxwell–Wagner Relaxation Limits Zero-Bias Capacitive Memory Window Readout in Ferroelectric $\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2$ Capacitors

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**Abstract**—The zero-bias capacitive memory window (CMW) enables non-destructive readout in ferroelectric capacitors, but its practical bandwidth is not set by polarization alone. We compare two  $\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2$  capacitor stacks by frequency-resolved  $C$ – $V$  spectroscopy from 1 kHz to 4 MHz together with PUND and switching-current measurements at 300 K and 10 K. The CMW of  $\text{TaN}/\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2/\text{TiN}$  relaxes as a single Debye process with  $f_c = 179$  kHz, while a  $\text{WO}_x$ -bearing stack with nearly the same low-frequency dielectric ceiling relaxes at  $f_c = 1.4$  kHz. At 10 K the capacitive read window falls below 1 kHz even though polarization switching persists. The data support an interface-limited Maxwell–Wagner mechanism: polarization stores the bit, but interfacial conductivity sets whether and how fast the states remain capacitively distinguishable.

**Index Terms**—Ferroelectric capacitor, hafnium zirconium oxide, capacitive memory window, non-destructive readout, Maxwell–Wagner relaxation.

## I. INTRODUCTION

Ferroelectricity in doped  $\text{HfO}_2$  and especially in  $\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2$  has enabled scalable memory devices compatible with CMOS and BEOL integration [1]–[3]. For ferroelectric capacitors, however, the standard readout is destructive because recovering the stored state requires sensing the switched charge  $2P_r$ . A capacitive read avoids that penalty: the two remanent polarization states exhibit different small-signal permittivities, so a sub-coercive capacitance measurement can distinguish the stored bit without forcing a rewrite [4].

Recent work has shown that this capacitive memory window can be enlarged and even used for array-level non-destructive readout, but the reported values are usually quoted at a single frequency near room temperature [5], [6]. That leaves the controlling physics unresolved. A finite zero-bias window cannot follow from polarization magnitude alone, because an ideally symmetric stack would still make the two  $C$ – $V$  branches intersect at zero bias. The missing question is therefore not whether polarization is present, but what sets the bandwidth, temperature range, and portability of the zero-bias read margin across device stacks.

Here we test two competing pictures. If the CMW is ferroelectric-limited, its dynamics should track the intrinsic dielectric and switching response of the  $\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2$  layer. If it is interface-limited, its characteristic time should instead follow the interfacial impedance and change strongly with

stack design and temperature. The reduced manuscript is organized around that question and the answer it supports: polarization stores the bit, but the interface sets whether and how fast the two states remain capacitively distinguishable.

## II. EXPERIMENTAL METHODS

Two metal–ferroelectric–metal stacks were compared: Stack A,  $\text{TaN}/\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2(10\text{ nm})/\text{TiN}$ , and Stack B,  $\text{TiN}/\text{PE-TiN}/\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2/\text{WO}_x/\text{W}$ , where the latter introduces a deliberate bottom interfacial layer [7]. Grazing-incidence XRD was used to confirm crystallization, and frequency-resolved  $C$ – $V$  measurements were acquired in parallel  $C_p$ – $R_p$  mode with a 50 mV rms probe from 1 kHz to 4 MHz. The dc bias was swept  $-2 \rightarrow +2 \rightarrow -2$  V so that the two polarization branches were recorded under identical conditions.

For each frequency, the zero-bias capacitive memory window was extracted from the difference between the branch permittivities,

$$\text{CMW}_\xi(f) = \xi_\downarrow(0, f) - \xi_\uparrow(0, f), \quad (1)$$

where  $\xi_\uparrow$  and  $\xi_\downarrow$  are the branches reached after opposite poling. PUND and switching-current measurements were then used to verify that the polarization state remained switchable independently of the capacitive readout. The temperature dependence was tested by repeating the electrical measurements at 300 K and 10 K with the same probe and instrument configuration. Main-text interpretation was restricted to frequencies below the series-resistance-limited regime identified from the high-frequency roll-off [8], [9].

## III. RESULTS AND DISCUSSION

At 1 kHz, both stacks show the butterfly-shaped permittivity expected for ferroelectric  $\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2$ , but they do not exhibit the same zero-bias read margin. Stack A gives  $\text{CMW}_\xi \approx 5.2$ , whereas Stack B gives  $\text{CMW}_\xi \approx 2.3$  at a similar peak permittivity of about 62 [Fig. 2]. As frequency increases, the branch separation collapses even though a substantial dielectric response remains. The key observation is that the state-dependent part disappears faster than the underlying permittivity, so the read margin is carried by a dispersive contribution rather than by the full ferroelectric response.

The frequency dependence of the window is well described by a single-Debye relaxation,

$$\text{CMW}(f) = \frac{\text{CMW}_0}{1 + (f/f_c)^2}. \quad (2)$$

For Stack A the fit gives  $\text{CMW}_0 = 5.38$  and  $f_c = 179$  kHz ( $\tau = 0.90 \mu\text{s}$ ), while Stack B relaxes at  $f_c = 1.4$  kHz ( $\tau = 114 \mu\text{s}$ ) [Fig. 3]. Differential loss and the Cole–Cole analysis reported in the full manuscript support the same single-process interpretation [10], [11]. This argues against a broad ferroelectric relaxation spectrum as the dominant origin of the zero-bias CMW.

Temperature further separates capacitive readout from switching. At 10 K, Stack A still switches, with  $2P_r$  reduced from 33.1 to 20.3  $\mu\text{C}/\text{cm}^2$  and  $E_c$  increased from 0.93 to 1.17 MV/cm, but the capacitive read window shrinks to about 2.2 at 1 kHz and falls to the noise floor above 1.1 kHz [Fig. 4]. Cooling therefore suppresses the CMW bandwidth by more than two decades while leaving polarization reversal measurable, which is not what a ferroelectric-limited readout would predict [12], [13].

The strongest evidence for an interface-controlled clock comes from comparing the two stacks. Their low-frequency dielectric ceilings are nearly identical, yet their relaxation times differ by  $126\times$  [Table I and Fig. 4]. A compact Maxwell–Wagner description,

$$\tau \approx \frac{\varepsilon_i}{\sigma_i}, \quad (3)$$

accounts for that split by assigning the timescale to a more resistive interfacial layer rather than to the ferroelectric bulk [14], [15]. The extracted interfacial areal capacitances differ only modestly (2.95 versus 2.34  $\mu\text{F}/\text{cm}^2$ ), whereas the corresponding areal resistances differ by about  $157\times$ . The usable zero-bias NDRO band is therefore an interface property, and single-frequency CMW values are not portable across stacks unless the corner frequency is also reported.

#### IV. CONCLUSION

The conference version reduces the manuscript to one governing result: zero-bias capacitive readout in ferroelectric  $\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2$  capacitors is limited by an interfacial Maxwell–Wagner relaxation rather than by the ferroelectric’s intrinsic switching timescale. The CMW collapses with frequency while a substantial dielectric response remains, its relaxation follows a single-Debye law, and cooling suppresses the read bandwidth far more strongly than switching itself. Because two stacks with nearly identical dielectric ceilings differ by more than two orders of magnitude in relaxation time, the usable NDRO read band is an interface property. Reported CMW values must therefore be tied to measurement frequency and stack design.

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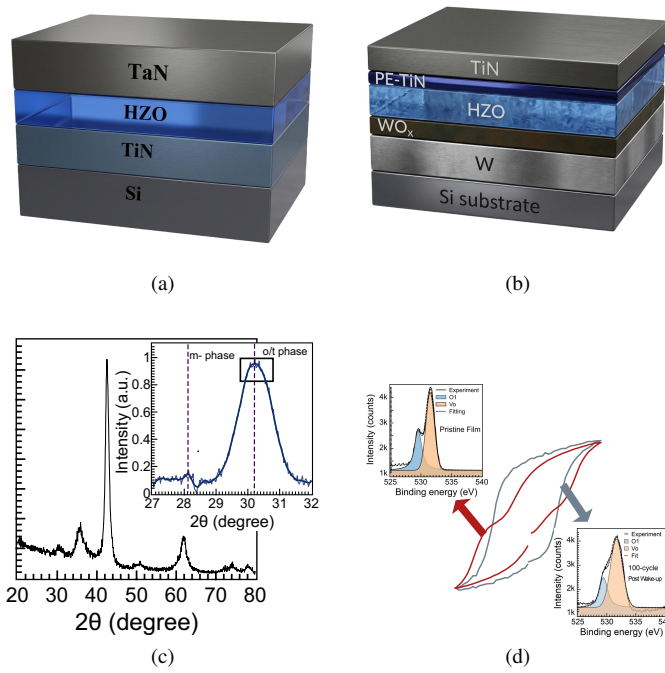


Fig. 1. Device and materials context. (a), (b) The compared capacitor stacks differ mainly in their electrode/interface design. (c) GIXRD confirms ferroelectric-phase formation in the  $\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2$  films. (d) Polarization and interfacial-chemistry context from the source manuscript motivate the interface-focused interpretation used in this four-page version.

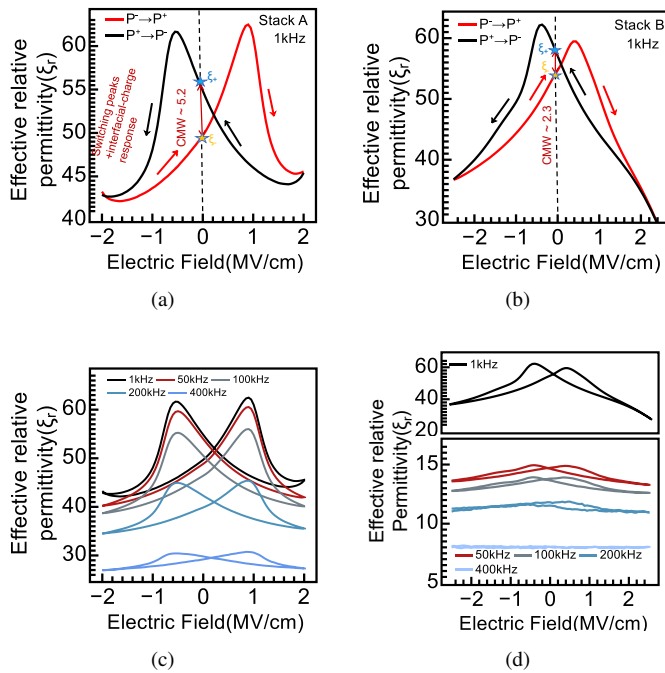


Fig. 2. Definition and collapse of the zero-bias CMW. Stack A shows  $\text{CMW}_\xi \approx 5.2$  at 1 kHz, while Stack B shows  $\approx 2.3$  at a similar peak permittivity. With increasing frequency, the branch separation collapses while a substantial dielectric response remains, showing that the lost quantity is the state discrimination rather than the full capacitance.

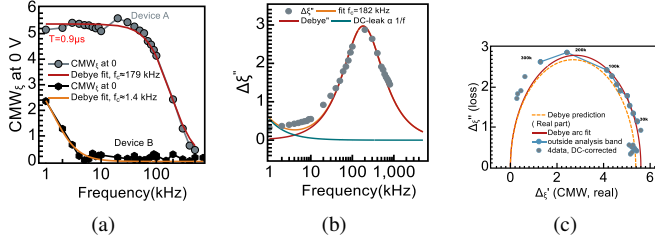


Fig. 3. Single-process relaxation evidence. The CMW roll-off, differential loss, and Cole-Cole response are all consistent with a single-Debye relaxation, giving  $f_c = 179$  kHz for Stack A and  $f_c = 1.4$  kHz for Stack B.

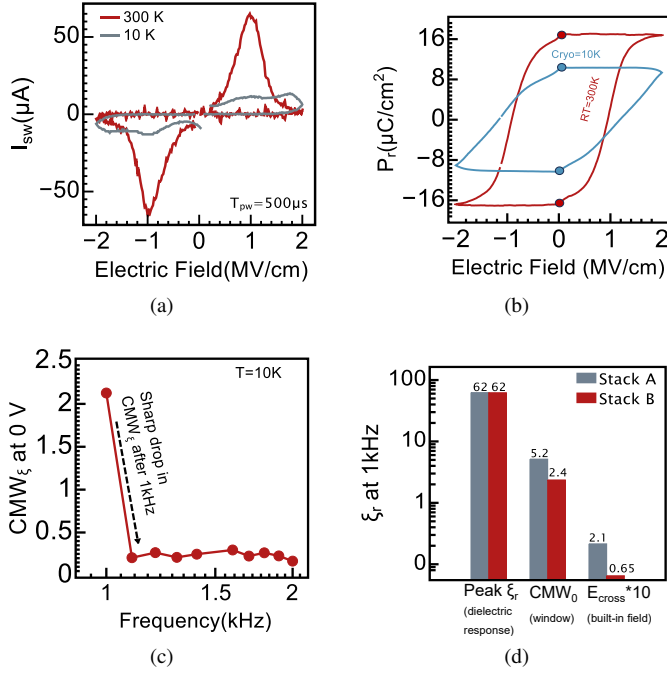


Fig. 4. Temperature and interface summary. At 10 K, switching persists but the zero-bias CMW bandwidth collapses below 1 kHz. The cross-stack summary shows nearly identical low-frequency dielectric ceilings together with large differences in CMW amplitude and clock speed, consistent with interface-controlled readout.

TABLE I  
KEY PARAMETERS RETAINED FOR THE 4-PAGE VERSION

| Parameter              | Stack A                              | Stack B                        |
|------------------------|--------------------------------------|--------------------------------|
| Peak $\xi_r$ at 1 kHz  | 62.5                                 | 62.2                           |
| Ceiling $\xi_r(0)$     | $62.4 \pm 1.2$                       | $62.7 \pm 0.9$                 |
| CMW at 1 kHz           | 5.2                                  | 2.4                            |
| CMW <sub>0</sub> (fit) | 5.38                                 | 3.6                            |
| $f_c$                  | 179 kHz                              | 1.4 kHz                        |
| $\tau$                 | 0.90 $\mu$ s                         | 114 $\mu$ s                    |
| $E_{cross}$            | 0.221 MV/cm                          | 0.081 MV/cm                    |
| $c_{IL}$               | 2.95 $\mu$ F/cm <sup>2</sup>         | 2.34 $\mu$ F/cm <sup>2</sup>   |
| $R_{IL}$               | 0.31 $\Omega$ -cm <sup>2</sup>       | 48.8 $\Omega$ -cm <sup>2</sup> |
| $2P_r$ at 300 K / 10 K | 33.1 / 20.3 $\mu$ C/cm <sup>2</sup>  |                                |
| CMW at 300 K / 10 K    | 5.2 / 2.2 (Stack A)                  |                                |
| $f_c$ at 300 K / 10 K  | 179 kHz / $\lesssim 1$ kHz (Stack A) |                                |